



Threat Model Comparison

How four architectures respond to the same ten attacks

ZeroPort

Zero ports. Zero trust. Zero blind spots.

	Fully exposed	Partially mitigated	Structurally prevented	
	Traditional VPN + Firewall	Commercial ZTNA (vendor cloud)	Port-knocking / SPA	ZeroPort
Internet port scan	Answers, fingerprinted	Endpoint discoverable	Listener still answers	Nothing responds
DDoS on the entry point	Concentrator saturates	Vendor edge chokepoint	Knock listener flooded	No entry point to flood
Zero-day in the gateway	Full perimeter breach	Vendor patch cycle	Knock parser exploitable	No exposed daemon
Stolen / lost device	Valid until revoked	Needs reachability	Secret still valid	Lease self-expires
Rogue or coerced admin	One admin = full access	One console admin	One operator holds key	2-of-3 quorum required
Insider lateral movement	Flat once inside	Vendor-enforced policy	Full access after knock	Per-port segmentation
Audit-log tampering	Root can edit the file	Vendor-held, unverifiable	Minimal logging	Chained + anchored
Attribution of a flow	IP-to-IP guesswork	Known to the vendor	None	Cryptographic identity
Sovereignty / foreign dep.	Vendor firmware	Foreign cloud control	Self-hosted	Relays in-Kingdom
Metadata to an observer	Static public gateway	Vendor sees metadata	Peer IPs visible	Per-flow: Tier 1 dark
STRUCTURALLY PREVENTED	0 / 10	0 / 10	1 / 10	9 / 10

The perimeter protects the network by exposing it. Commercial ZTNA removes the port but relocates the trust into a cloud you cannot audit. SPA still runs a listener. ZeroPort removes the listening surface, distributes the authority, and makes the audit independently verifiable.

One cell in our column is amber on purple: direct tunnels reveal peer IPs, so sensitive flows are pinned to the onion path. We show the trade rather than hide it.

<https://zeroport.vercel.app/>